Project Plan

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Motivation

The motivation behind this project is to develop a model that can assess speech quality in a descriptive string, similar to an audio expert. Demonstrating this as a proof of concept — that AI can evaluate audio in a comparable manner to a human expert — could significantly reduce the cost of large-scale data labeling.

Current SOTA audio-assessment datasets like NISQA contain MOS (Mean Opinion Scores) and other float values assessing quality; however, no current dataset exists that includes a text label describing the quality.
Filling this gap by creating such a dataset could potentially support research in noise-cancelling and hearing aid development.

Research Question

Is it possible to train an audio LLM so that, given human audio as input, it outputs a high-quality audio assessment?

Our hypothesis is that this is possible. Step one of this project will be to re-implement the paper "Audio Large Language Models Can Be Descriptive Speech Quality Evaluator"[1]. From this standpoint, we will attempt to improve the model so that, instead of generating a "generic" string, it outputs something similar to an audio experts, in relation to GN's inquiry.

Answering this research question could contribute new knowledge by bridging the gap between numerical quality scores and expert-level descriptive evaluations.

Time plan

In Figure 1 is our Gantt chart. We will update it dynamically, and you have access to our daily meetings and Gantt chart here: Project Gantt Chart and Meeting Notes

Risk analysis of possible delays:

- Re-implementing: The original paper does not include source code, but we do have access to their datasets and Swift scripts. Re-implementing in Python for more control over our experiment may be challenging and time-consuming.
- Analyzing results & continuing development: Historically, we have experienced delays when finalizing code and moving forward instead of making additional changes. This typically occurs after analyzing results and identifying mistakes or potential improvements.

Task	Status	Due Date	Notes
Literature Review	Complete	Mar 26, 2026	
Define Project Scope & Goals	Complete	Feb 3, 2026	
Develop Project Plan	In Progress	4. mar. 2026	
Reimplement Paper Method	In Progress	Mar 11, 2026	
Finalize Model Scope	To Do	Mar 23, 2026	Setup Meeting
Implement Method Changes	To Do	May 31, 2026	
Analyze Results	To Do	May 31, 2026	
No More Coding!	To Do	May 31, 2026	
Report Writing	To Do	Jun 23, 2026	
New Discussion Methods	To Do	Jun 23, 2026	Small implementations of new ideas for improvement
Proofreading	To Do	Jun 26, 2026	
Hand In	To Do	Jun 26, 2026	
Prepare Final Presentation	To Do	25. jun. 2026	Depends On Exam Date

Figure 1: Gantt chart of the project plan.

References

[1] Chen Chen, Yuchen Hu, Siyin Wang, Helin Wang, Zhehuai Chen, Chao Zhang, Chao-Han Huck Yang, and Eng Siong Chng. Audio large language models can be descriptive speech quality evaluators. In *International Conference on Learning Representations (ICLR) 2025*, 2025.